

Bachelor of Science (Honours) in Data Science and Artificial Intelligence

DA 102 Data Analysis Basics



Introduction



Learning Objectives



- 01** Know various types of data
- 02** Issues associated with data
- 03** Understand broad group of analytics
- 04** Understand what is descriptive analytics
- 05** Understand what is predictive analytics
- 06** Tools used to perform analytics





About Data



Data



- **Definition: collection of information, facts, or values**
- **The data is used as basis for computation, analysis and decision making.**
- **Data can be in the form of numbers, text, images, audio, video.**
- **Data is the raw material that is processed, organized, interpreted to extract meaning and generate insights.**
- **Data is a fundamental concept that plays a crucial role in various fields such as science, business, and research.**
- **Data is considered as the key in this era of digital age**

Examples of data

Calibri (Body) 11				
C3			4.3	
	A	B	C	D
1	Roll Number	CPI	Feedback	
2	230101001	7.1	4.1	
3	230101002	7.2	4.3	
4	230101003	8.3	4.7	
5	230101004	8.7	4.9	
6	230101005	8.6	4.9	
7	230101006	9.7	5	
8	230101007	9.1	5	
9	230101008	8.8	4.9	
10	230101009	8.8	4.9	
11	230101010	7.7	4.5	
12	230101011	7.3	4.3	
13	230101012	7.1	4.1	
14	230101013	6.5	4.9	
15	230101014	4.7	3	
16	230101015	9.3	4.5	
17				



JYSK UK
11 Oct 2021 · 🌐

For the first time ever in JYSK Ireland & UK history we gathered all our Senior Leaders and Store Managers together in person to look back on the last year and of course celebrate in style with our incredible JYSK Awards Evening .

We know the last few years have been a challenge for everyone but with the ambition, attitude and corporate spirit we saw this week, we are sure our teams are ready to take on the busy months ahead.

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#gettogether #bringdedicationmeetpossibilities



👍❤️ 40

10 comments 12 shares



Post



india leading the world



935



Rajesh @RajeshShares · 3m
Mirchi lagi ?



5



Albert J Khare @albertkhare1981 · 1h

I hardly use cash nowadays, even the smallest vendors have UPI
The smallest transaction I made using UPI was 10/- but I have seen Rs 3/- being paid as well.
Might be a good time to introduce a digital currency which is not linked to INR but rates set by RBI



5

1,425



Lala @FabulasGuy · 1h
मिर्ची स्पेशल @PChidambaram_IN के लिए



Examples of data



TOI Explainers Videos City India World Business Tech Cricket Sports Entertainment Auto TV Web Series Life & Style Education

China has taken over our grazing land, says Rahul...

Should you be worried about new Covid virus...

Some members left NCP due to ED probe: Shar...

2nd T20I Live: Ruturaj solid despite...

German embassy praises UPI, shares video of...

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'Was a little careful in...', says Jasprit Bumrah

'Made-in-India'

'Can't sleep at

Amid 'Gadar 2' success, Sunny Deol lands in...

Swara Bhasker indirectly criticises Alia Bhatt for...

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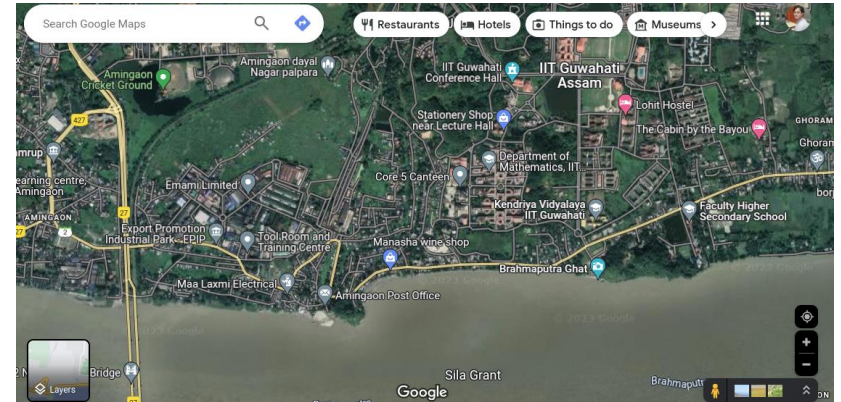
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WHY RUSSIA'S LUNA-25

RUSSIA'S LUNA-25

INDIA AND RUSSIA RECON



Examples of data



Today Internet		
Arrival	Day 1 - Mar 14, Thu	Departure
9:36 AM 9:46 AM	Ponduru 0 km Platform 3	9:38 AM 9:46 AM
9:46 AM 9:54 AM	Dusi 9 2 km to Srikakulam Road (Updated few seconds ago)	9:46 AM 9:54 AM
9:53 AM 10:01 AM	S 15 km Platform 2	9:55 AM 10:02 AM
10:12 AM 10:14 AM	Tilaru 35 km Platform 2	10:13 AM 10:15 AM
10:26 AM 10:26 AM	Kotabommali 48 km Platform 1	10:28 AM 10:28 AM
10:41 AM 10:41 AM	Naupada Junction 62 km Platform 2	10:43 AM 10:43 AM
17016-Visakha Express (PSA-BBS)		





About Data



Data 01



Numbers

Quantity: 10

Price: 347.73

Qualitative Data

{"Un-married",
"Married", Divorced"}

{"Student", "Faculty",
"Staff"}

Quantitative Data

{1 = Poor, 2 = Fair, 3 =
Good, 4 = Very Good,
5 = Excellent}

{AA = 10, AB = 9, BB =
8, ..., DD = 4, F = 0}

Boolean Values

{True, False}

Data 02



Temporal Data

Date: 15-Aug-2023

Time: 11:30 AM

**Time Stamp:
11:30 AM, 15-Aug-2023**

Text Data

**Sequence of characters such as
words, sentences or paragraphs**

Mixed data

**Combination of any of
these data types**

Data 03



Audio

**An encoded file
corresponding to a song**

Image Data

Data captured through cameras

**Data captured through
screenshots in computers**

Video data

**Combination several
images and an
associated audio data**



Data – stored in spread sheet



Excel example-data

Search (Alt + Q)

File Home Insert Draw Page Layout Formulas Data Review View Help

Comments Catch up Editing Share

Calibri (Body) 11 B

General

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Student Name	Semester	Gender	Hostel Name	Hostel Room No	CPI	Result									
2	Plato Macias	2	M	Manas	224	1.08	Fail									
3	Neve Wilder	5	F	Manas	128	10	Pass									
4	Mallory Cohen	3	F	Lohit	322	6.5	Pass									
5	Preston Houston	7	M	Brahmaputra	221	6.5	Pass									
6	Dorian Sanford	8	M	Umiyam	320	8.67	Pass									
7	Jullen Castaned	6	M	Dihing	221	5.42	Pass									
8	Allegra Vargas	6	M	Lohit	223	2.17	Fail									
9	Jesse Fry	7	F	Siang	230	9.75	Pass									
10	Cheryl Spencer	3	M	Lohit	226	3.25	Fail									
11	Wyoming Cobb	5	M	Dihing	227	9.75	Pass									
12	Declan Mathis	8	F	Dhansiri	121	10	Pass									
13	Curran Floyd	3	F	Dibang	323	6.5	Pass									
14	Hunter Camacho	6	M	Siang	222	10	Pass									
15	Quentin Simon	4	M	Kameng	326	6.5	Pass									
16	Liberty Cooke	4	F	Kapilli	123	4.33	Fail									
17	Lillian Mueller	2	F	Dihing	225	8.67	Pass									
18	Beck Molina	4	F	Umiyam	220	4.33	Pass									
19	Kalia Floyd	3	M	Umiyam	321	1.08	Fail									
20	Luke Finch	1	M	Lohit	322	1.08	Fail									
21	Ria Warner	8	F	Dibang	325	3.25	Fail									
22	Ivana Mendoza	4	M	Siang	129	10	Pass									
23	Althea McMahon	3	M	Dibang	221	8.67	Pass									
24	Ainsley Harper	2	M	Dibang	222	5.42	Pass									

Sheet1

Calculation Mode: Automatic Workbook Statistics

Give Feedback to Microsoft 100%

Data – stored in a note pad



```
example-data.csv
~/Downloads

Student Name,Semester,Gender,Hostel Name,Hostel Room No,CPI,Result
Plato Macias,2,M,Manas,224,1.08,Fail
Neve Wilder,5,F,Manas,128,10,Pass
Mallory Cohen,3,F,Lohit,322,6.5,Pass
Preston Houston,7,M,Brahmaputra,221,6.5,Pass
Dorian Sanford,8,M,Umiam,320,8.67,Pass
Cullen Castaneda,6,M,Dihing,221,5.42,Pass
Allegra Vargas,6,M,Lohit,223,2.17,Fail
Jesse Fry,7,F,Siang,230,9.75,Pass
Cheryl Spencer,3,M,Lohit,226,3.25,Fail
Wyoming Cobb,5,M,Dihing,227,9.75,Pass
Declan Mathis,8,F,Dhansiri,121,10,Pass
Curran Floyd,3,F,Dibang,323,6.5,Pass
Hunter Camacho,6,M,Siang,222,10,Pass
Quentin Simon,4,M,Kameng,326,6.5,Pass
Liberty Cooke,4,F,Kapilli,123,4.33,Fail
Lillian Mueller,2,F,Dihing,225,8.67,Pass
Beck Molina,4,F,Umiam,220,4.33,Pass
Kalia Floyd,3,M,Umiam,321,1.08,Fail
Luke Finch,1,M,Lohit,322,1.08,Fail
Ria Warner,8,F,Dibang,325,3.25,Fail
Ivana Mendoza,4,M,Siang,129,10,Pass
Althea McMahon,3,M,Dibang,221,8.67,Pass
Ainsley Harper,3,M,Disang,222,5.42,Pass
Chaney Durham,7,M,Dhansiri,229,3.25,Fail
Kelly Hill,3,F,Siang,130,4.33,Pass
Allegra Maynard,1,F,Kameng,226,7.58,Pass
Ethan Rivera,7,M,Dihing,328,8.67,Pass
Gay Maddox,3,M,Dihing,124,6.5,Pass
Georgia Downs,6,F,Kameng,225,7.58,Pass
William Beck,3,F,Kameng,125,1.08,Fail
Sebastian Potter,3,M,Dihing,329,7.58,Pass
Aileen Reeves,6,M,Manas,322,9.75,Pass
George Schroeder,2,M,Kameng,325,8.67,Pass
Hedda Howell,7,M,Disang,322,8.67,Pass
```

CSV Tab Width: 8 Ln 1, Col 1 INS





Real valued data



Analysis - example 01



- **To buy the book titled "The Linux Programming Interface" by Michale Kerrisk from online website**
- **The objective is to buy the above book at a website which offers best price.**
- **To achieve this objective, collect prices from various ecommerce websites**

When data is real valued



Analysis – example 01



amazon.in Deliver to Dr Guwahati 781039 All the linux programming interface

Books Advanced Search New Releases & Pre-orders Best Sellers Browse Genres Children's & Young Adult Textbooks Exam Central All Indian Languages

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THE LINUX PROGRAMMING INTERFACE
A Linux and UNIX System Programming Handbook
MICHAEL KERRISK

The Linux Programming Interface: A Linux and UNIX System Programming Handbook Hardcover – 1
October 2010
by Michael Kerrisk (Author)
4.8 stars 619 ratings See all formats and editions

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When data is real valued

Analysis – example 01



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THE LINUX PROGRAMMING INTERFACE
A Linux and UNIX* System Programming Handbook
MICHAEL KERRISK

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Delivery by 19 Aug, Saturday | **Free ₹40**

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When data is real valued

Analysis – example 01



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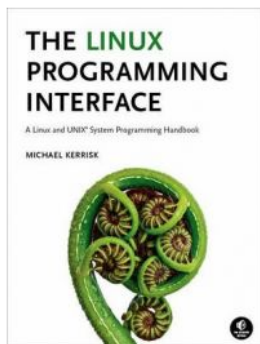
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by M. Kerrisk

Format: Hback

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Analysis – example 01



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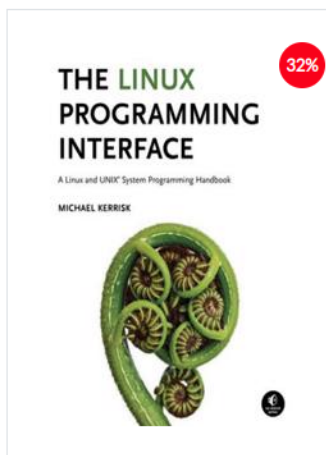


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The Linux Programming Interface: A Linux and UNIX System Programming Handbook

(Hardcover) | Released: 01/01/0001

| Publisher: No Starch Press

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When data is real valued

Analysis – example 01



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- Medical Books 1
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Price

Rs. 782 - Rs. 810

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Customer Rating

★★★★★ & Up 1

We've got 2 results for "the linux programming interface"

Sort by: Relevance

Enter your pincode Check

4 left!

The Intel Microprocessors, Architecture, Programming, and Interfacing

1 left!

The 8088 and 8086 Microprocessors

Programming, Interfacing, Software, Hardware and Applications

Waiting for googleads.g.doubleclick.net...

When data is real valued

Analysis – example 01



Website	Price	Additional information
www.amazon.in	5594	In stock
www.flipkart.com	5599	In stock
books.rediff.com	5378	Out of stock
bookswagon.com	7738	Out of stock
www.snapdeal.com	Not available	Not available

When data is real valued

Analysis – example 01



Website	Price	Additional information
www.amazon.in	5594	In stock
www.flipkart.com	5599	In stock
books.rediff.com	5378	Out of stock
bookswagon.com	7738	Out of stock
www.snapdeal.com	Not available	Not available

- Prices are real valued numbers
- Price at snapdeal.com is not available due to unavailability of product
- Computing lowest price from the second column of the table is a challenging task as price information is unavailable for all websites

When data is real valued

Analysis – example 01



Website	Price	Additional information
www.amazon.in	5594	In stock
www.flipkart.com	5599	In stock
books.rediff.com	5378	Out of stock
bookswagon.com	7738	Out of stock
www.snapdeal.com	0	Not available

When data is real valued

- The price offered by snapdeal is made 0 to make price a real valued number.
- However, this change leads to factually incorrect information that the lowest price is offered snapdeal.
- Handling "not available" values is very important in data analysis otherwise, decisions swing significantly resulting in errors

Analysis – example 01



Website	Price	Additional information
www.amazon.in	5594	In stock
www.flipkart.com	5599	In stock
books.rediff.com	5378	Out of stock
bookswagon.com	7738	Out of stock
www.snapdeal.com	Not available	Not available

When data is real valued

- Lowest price is offered by books.rediff.com
- Additional information tells us that though this is the best price offered, the book is out of stock.
- We therefore must search for second lowest price.
- Sort price in descending order and pick the second element in sorted list.
- We understand that Amazon Offers the second lowest price.



Analysis – example 01



Website	Price	Additional information
www.amazon.in	5594	In stock
www.flipkart.com	5599	In stock
books.rediff.com	5378	Out of stock
bookswagon.com	7738	Out of stock
www.snapdeal.com	Not available	Not available

When data is real valued

- Sampling – 5 websites are visited of many e-commerce websites.
- Computing minimum price value lead to decision making
- Is this example too easy?
Well the fact is:
- 9 out of 10 consumers price check a product on Amazon
- <https://www.bigcommerce.com/blog/amazon-statistics/>





Quantitative data



Analysis – example 02



novation The Essential Keyboard Controller ₹29,744⁰⁰ prime

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Site Reliability Engineering: How Google Runs Production Systems 1st Edition, Kindle Edition

by Niall Richard Murphy (Author), Betsy Beyer (Author), Chris Jones (Author), & 1 More | Format: Kindle Edition

4.6 ★★★★★ 1,043 ratings See all formats and editions

4.6 out of 5
1,043 global ratings

Star Rating	Percentage
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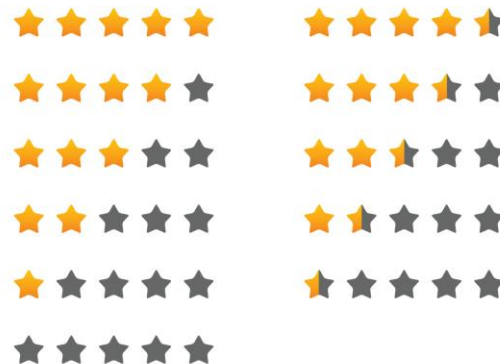
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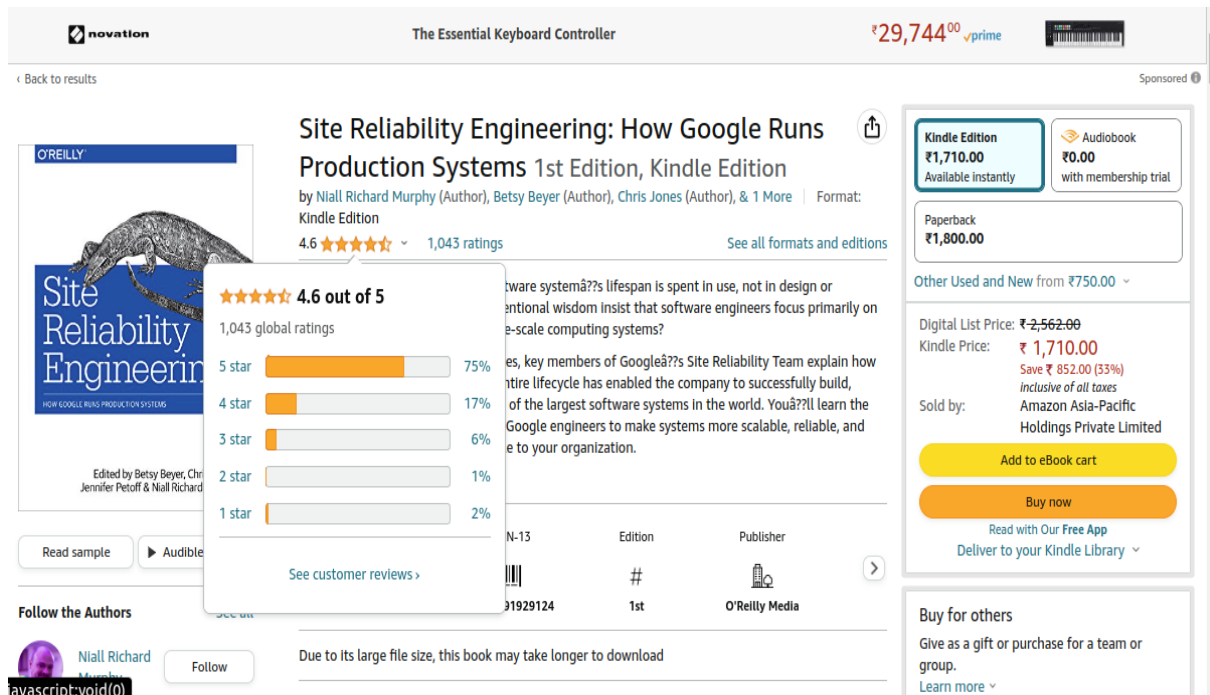
When data is Quantitative

- Visit www.amazon.com
- Search for any product and access associated product description page.
- Apart from book description, we get to see ratings given by users
- The ratings are quantitative values such as



<https://www.vecteezy.com/>

Analysis – example 02



- A total of 1043 users gave ratings
- 75% of users gave 5 stars
- 17% of users gave 4 stars and so on
- The ordered values are presented as histogram visualization
- Average rating is computed and presented as 4.6 out of 5 stars

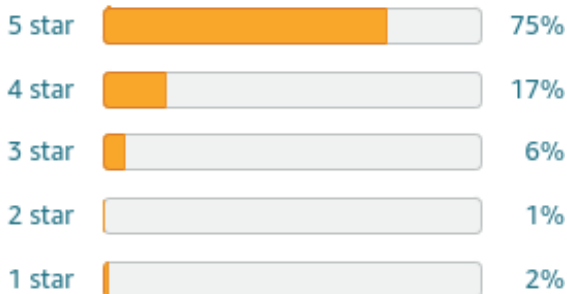
When data is Quantitative

Analysis – example 02



★★★★☆ 4.6 out of 5

1,043 global ratings



[See customer reviews >](#)

User 1	5 stars
User 2	5 stars
User 3	4 stars
User 4	1 star
...	...
User 1043	5 stars

- **Analysis: Computed the following**
- **Number of users who gave 5 stars**
- **Number of users who gave 4 starts**
- ...
- **Number of users who gave 1 star**
- **Average rating across all users**

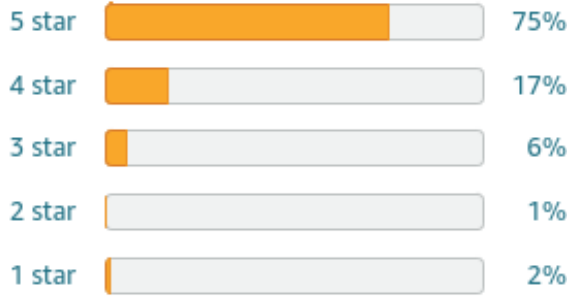
When data is Quantitative

Analysis – example 02



★★★★☆ 4.6 out of 5

1,043 global ratings



[See customer reviews >](#)

- **Product ratings influence in conversions from viewing to buying**
- **A correlation between ratings and conversion has been observed – a product with 3.7 rating has 15% more click through rate (CTR)**
- **Analysis involving only ratings**

When data is Quantitative

Analysis – example 02



User 1	5 stars
User 2	5 stars
User 3	4 stars
User 4	1 star
...	...
User 1043	5 stars

- **From the data perspective**
- **The rating data may be visualized to be stored as shown**
- **When users visits product page, average rating and histograms are generated from this kind of data.**
- **Analytics play a pivotal role in transforming data and influencing decisions**

When data is Quantitative







Qualitative data



Analysis – example 03



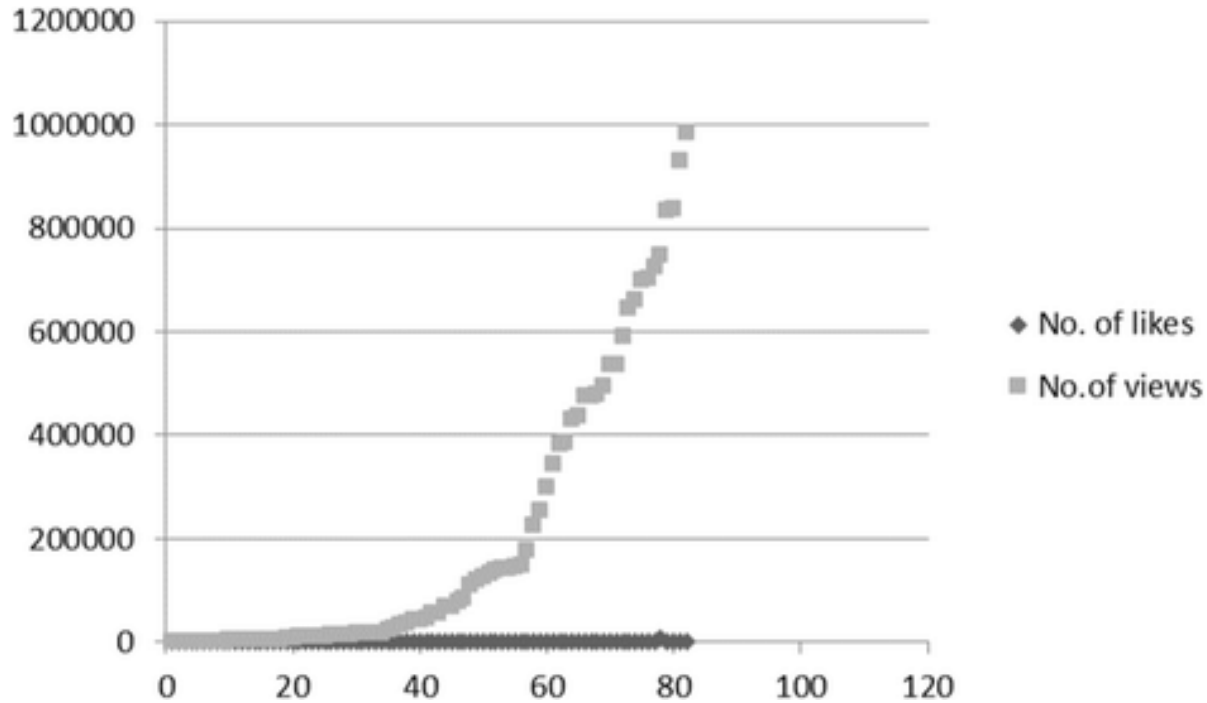
- Visit www.youtube.com
- Search for any video
- Apart from the video, we get to see how many users {liked, disliked} the video
- More likes of a video has strong correlation to increase in revenue



<https://www.vecteezy.com/>

When data is Qualitative

Analysis – example 03



- Visit www.youtube.com
- Search for any video
- Apart from the video, we get to see how many users {liked, disliked} the video
- More likes of a video has strong correlation to increase in revenue



<https://www.vecteezy.com/>

When data is Qualitative





Text data



Analysis – example 04



Nikhil_CV

★★★★★ **Learn from the mistakes of a leader**

Reviewed in India on 27 January 2021

Verified Purchase

It is always a good idea to learn from what a great organization like Google has to tell us. This book will be handy to anyone who needs to know what SRE and DevOps world looks like. I wish this was an elective in Indian colleges for IT/CS or any computer streams

The book is printed B&W in normal A4 like sheet and bound together and cut. Exactly felt like an academic book in look and feel, but the contents are exciting (might be a bit dated, never out-dated though)

I would suggest not to buy this for long term storage as the print ink may leech to the other side of the page if overexposed. Cheap material but great content!

3 people found this helpful

Helpful

| Report

- Visit www.amazon.com
- Search for any product and access associated product description page.
- Apart from book description, we get to read detailed product review given by customers who bought the product
- This information is in the text form.

Analysis – example 04



Chris B.



Useful But Needlessly Verbose

Reviewed in the United States on 7 January 2023

Verified Purchase

I enjoyed reading this book and learning about challenges Google faced internally as they grew their SRE team. I found some of the "war stories" about big outages fun to read, like Google Music having to restore customers' music data from tape drives via trucks.

The most useful parts of the book to me were around scaling the SRE team and how they went from white glove support, to identifying and recommending best practices, to finally building out automated tools to provide teams with these best practices.

Some chapters were pretty weak and could have been cut. Like a chapter near the end comparing Google SRE practices with outside companies basically amounted to "Google already does all the best practices that the industry uses." Other chapters just mostly repeated content contained elsewhere in the book.

Also, what about some open ended content/questions about further improvements that could be made?
Not present

2 people found this helpful

Report

- **Analyze the contents for finding out the sentiment of the customer**
- **In the analytics terms, perform sentiment analysis of customer reviews**
- **If the complete review is in positive sentiment, then the it suggests that customer is happy with the product.**



A computer program



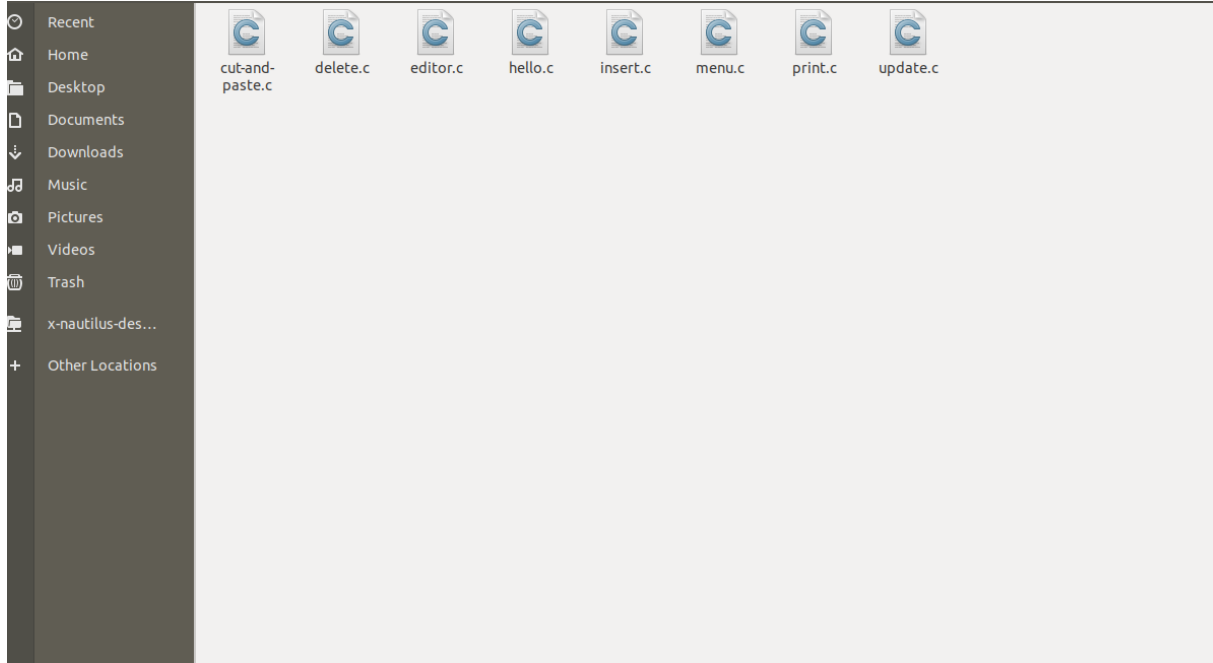
```
#include<stdio.h>
#include<stdlib.h>

int main(int argc, char *argv[])
{
    printf("Hello DA 102 learners, Welcome!\n");

    return EXIT_SUCCESS;
}
```

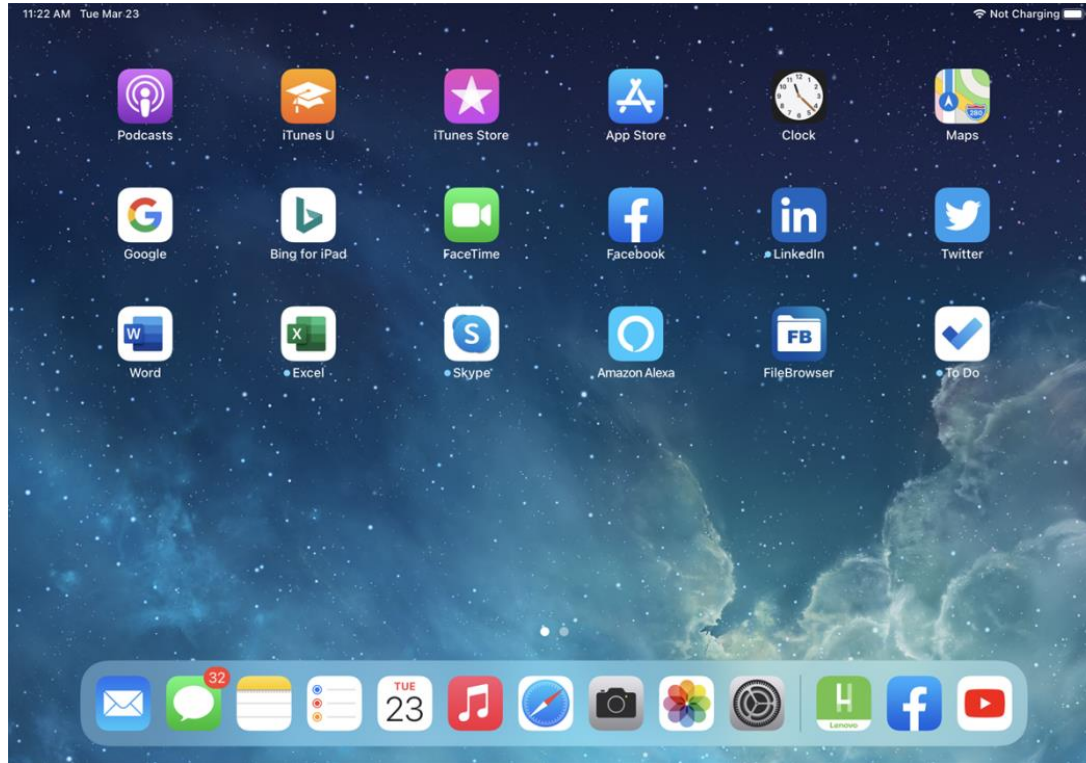
- A computer program is data
- Collection of such programs
- Collection of executables

Collection of programs



- **A computer program is data**
- **Collection of such programs**
- **Collection of executables**

Collection of executables



- **A computer program is data**
- **Collection of such programs**
- **Collection of executables**





Issues in Data Collection





Bias and Sampling Issues

- **Sampling Bias** - When the sample collected does not represent the entire population
- **Selection Bias** - When participants are not selected randomly, leading to skewed or inaccurate results
- **Nonresponse Bias** - When a significant portion of the selected participants does not respond, leading to a biased sample



Measurement Issues

- **Measurement Error:** Inaccuracies or inconsistencies in measurement instruments, leading to incorrect or imprecise data.
- **Subjective Measurements:** When data is collected through subjective methods like surveys, individuals might respond based on personal biases or interpretations.
- **Social Desirability Bias:** Participants may provide responses they believe are socially acceptable rather than their true opinions or behaviors



Data Quality and Integrity Issues

- **Data Integrity Issues**
- **Missing values**
- **A data record that is not complete.**
- **Data entry errors.**
- **Duplications in data.**



Ethical and Privacy Concerns

- **Informed Consent:** Obtaining proper informed consent from participants, especially in sensitive or intrusive research.
- **Privacy Protection:** Ensuring that personally identifiable information is not leaked or misused.
- **Anonymity:** Striking a balance between collecting meaningful data and protecting participants' identities

Technological Challenges



- **Data Security:** Protecting data from breaches, leaks, or unauthorized access.
- **Data Storage:** Ensuring proper and secure storage of collected data.
- **Technical Glitches:** Issues with data collection tools, software, or hardware can affect data quality.

Cultural and Language Related Challenges



- **Cultural Bias:** Data collection instruments might be biased towards a particular culture or group.
- **Translation Issues:** Translating surveys or questionnaires into different languages can lead to discrepancies in meanings.



Resource Limitations

- **Financial Constraints:** Adequate data collection might require funding for tools, personnel, and infrastructure.
- **Time Constraints:** Rushed data collection might lead to errors or incomplete data.



Types of analytics



Analytics components



- A business context in order to take up data analytics.
- Two examples to support the business context.
- Pregnant women are likely to be price-insensitive.
- Their willingness to spend more is the potential for business.
- Baby products related market share is 38 billion dollar.
- Given a female customer predict if the customer is pregnant or not.
- In the business context, if the prediction is accurate, offer pregnant women the customized products and services.

Analytics components



- A business context in order to take up data analytics.
- Two examples to support the business context.
- In competitive markets such as telecommunication industry, customers abruptly leave the services of the service provide.
- Predicting which customer leaves the service before hand help address core concerns of the customers.
- In the business context, if the prediction is accurate, offer discounts to retain such customers.

Analytics types



- **Descriptive analytics**
- **Predictive analytics**



Descriptive analytics





Descriptive analytics

- In descriptive analytics the focus is on summarizing and interpreting historical data
- It gives insights into past events and trends
- Strives to provide a clear and concise representation of data in order to understand the data and aid in decision making
- Summarizing large volumes of data into manageable and interpretable forms
- Identify patterns and trends present in the historical data
- Visualize data as visualization make it easier to communicate complex data



Descriptive analytics - example

- An e-commerce company want to analyze the sales data for a particular product over the past year.
- It has data describing each sale such as: date of the sale, the product sold, the quantity, and the revenue generated.

Date of Sale	Product Sold	Quantity	Revenue Generated (\$)
1/5/2023	Product A	10	150
1/10/2023	Product B	5	75
2/3/2023	Product A	8	120
2/15/2023	Product C	12	180
3/2/2023	Product B	6	90
4/9/2023	Product A	15	225
5/20/2023	Product C	10	150
6/7/2023	Product B	7	105
7/14/2023	Product A	9	135
8/3/2023	Product C	11	165



Descriptive analytics - example

- **Descriptive analytics include**
- **Summarization: Computing average revenue for each product**
- **Summarization: What is the total revenue of each product**
- **Visualization: Plot sales of product A over timeline**
- **Visualization: Plot Quantity sold every month for product A**
- **Determine top selling products based on total revenue generated by each product**
- **Customer segmentation: grouping customers based on purchase frequency, total spending by customer, identify high value and low value customers**
- **Peak sales times: Identify days of the week when sales are highest**





Predictive analytics





Predictive analytics

- In predictive analytics the focus is on finding future trends given historical data and current data
- The goal is to build models that can learn from historical data.
- Use the learning to make predictions about future events (data).
- The models are trained on known outcomes and patterns.
- The trained models are employed to predict new or unseen data.



Example - customer churn





Example - customer churn

- **Percentage of customers who stopped using the product or service**
- **In competitive marketplace identifying potential customer who would discontinue their services is key.**
- **If known beforehand the intent of the customer that he/she would leave the services company may initiate retention plan. Which may include offering a coupon or providing discount for the previous three months service etc.**
- **The main challenge is no customer will explicitly state the reason leaving the services**
- **The reasons are to be understood from multiple data sources. Here is an example**

Example - customer churn



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Customer ID	Mobile number	Service	Usage	Month	Bill Amount	Bill Paid	Number of Complaints raised	Numer of Complaints resolved	Number of complaints unresolved	Rating on the service		
1234	1234567890	Prepaid	100 GB	Jan	500	Yes	1	1	0	5 stars		
1234	1234567890	Prepaid	90 GB	Feb	450	Yes	2	2	0	4 stars		
1234	1234567890	Prepaid	80 GB	Mar	400	Yes	3	3	0	3 stars		
1234	1234567890	Prepaid	70 GB	Apr	350	Yes	4	4	0	2 stars		
1234	1234567890	Prepaid	70 GB	May	350	Yes	5	2	3	2 stars		
1234	1234567890	Prepaid	70 GB	Jun	350	Yes	6	3	3	1 star		
1234	1234567890	Prepaid	60 GB	Jul	250	Yes	8	4	4	1 star		
1234	1234567890	Prepaid	50 GB	Aug	200	No	0	0	0	NA		
1234	1234567890	Prepaid	0 GB	Sept	250	No	0	0	0	NA		
1234	1234567890	Prepaid	0 GB	Oct	250	No	0	0	0	NA		
1234	1234567890	Prepaid	0 GB	Nov	250	No	0	0	0	NA		
1234	1234567890	Prepaid	0 GB	Dec	250	No	0	0	0	NA		
1234	1234567890	Prepaid	0 GB	Jan	250	No	0	0	0	NA		

Data from billing department

Data from service department

Example - customer churn

- **In this example a correlation exists between number of complaints raised, resolved, unresolved and rating.**
- **Obtain an elaborate data for all active customers for past four months a snapshot of the same is given below.**

File Home Insert Draw Page Layout Formulas Data Review View Help

Calibri (Body) 11 B [Font Color] [Text Background Color] [Bullets] [Link] [Table] General \$.00 →.00 ∑ Z ↕

B10 X ✓ fx 1234567892

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	Customer ID	Mobile number	Service	Usage	Month	Bill Amount	Bill Paid	Number of Complaints raised	Num of Complaints resolved	Number of complaints unresolved	Rating on the service			
1														
2	1234	1234567890	Prepaid	70 GB	May	350	Yes	5	2	3	2 stars			
3	1234	1234567890	Prepaid	70 GB	Jun	350	Yes	6	3	3	1 star			
4	1234	1234567890	Prepaid	60 GB	Jul	250	Yes	8	4	4	1 star			
5	1234	1234567890	Prepaid	50 GB	Aug	200	No	0	0	0	NA			
6	1235	1234567891	Prepaid	300 GB	May	600	Yes	1	0	1	NA			
7	1235	1234567891	Prepaid	330 GB	Jun	650	Yes	2	3	0	5 stars			
8	1235	1234567891	Prepaid	220 GB	Jul	480	Yes	4	3	1	3 stars			
9	1235	1234567891	Prepaid	200 GB	Aug	450	Yes	8	6	2	2 stars			
10	1236	1234567892	Prepaid	220 GB	May	400	Yes	1	1	0	5 stars			
11	1236	1234567892	Prepaid	230 GB	Jun	400	Yes	0	0	0	NA			
12	1236	1234567892	Prepaid	270 GB	Jul	500	Yes	2	2	0	5 stars			
13	1236	1234567892	Prepaid	300 GB	Aug	600	Yes	3	3	0	5 stars			
14														

< > ≡ Sheet1 Sheet2 Sheet3 **Sheet4** +

Calculation Mode: Automatic Workbook Statistics Average: 1234567892 Count: 4 Sum: 4938271568 Give Feedback to Microsoft



Customer churn

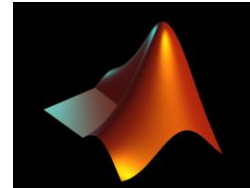
- When large number of customer data is presented, it is hard to establish relations between churn and the attributes of the data visually.
- When a relationship is established, the relationship is to be validated. In the previous example, customer ID 1235 is identified as potential to churn. This must be validated.
- The outline of predictive analytics is
- Perform data analysis and modeling: Analyze historical data and discover relationships, correlations, and patterns that can be used to make predictions. In the customer churn examples, complaints raised, number of unresolved complaints, rating on resolved complains are the information relationships.





Data analytics tools







Data analytics tools

- Several tools which are used for data analytics. Some of them are listed below
- **Microsoft Excel:** Widely used for basic data analysis, calculations, and charting.
- **Tableau:** Known for its powerful data visualization capabilities and interactive dashboards.
- **Power BI:** Microsoft's business analytics service for interactive visualizations and business intelligence.
- **QlikView/Qlik Sense:** Tools for data visualization, reporting, and business intelligence.
- **R:** A programming language and software environment for statistical computing and graphics.
- **Python:** A versatile programming language used for data analysis and manipulation with libraries like pandas and NumPy.



Data analytics tools

- **SAS (Statistical Analysis System):** Offers a suite of software for advanced analytics, business intelligence, and data management
- **SPSS (Statistical Package for Social Sciences):** Software for statistical analysis, used in social science research and data mining.
- **MATLAB:** A programming language and environment for numerical computing, often used in engineering and scientific research.
- **Stata:** A software package used for data analysis and statistical purposes
- **JMP:** Statistical discovery software often used for exploratory data analysis.



DA102 – Microsoft Excel



- This course focus on basics of data analysis and business modeling
- Microsoft Excel software tool is used.
- The choice of this tool stems from the fact that students of this online programme are in their first semester and are yet to introduce to programming languages.
- For those students who are introduced to the programming languages this course should enrich them with the orientation of data manipulation through software systems.



DA102 – Detailed contents

- **Basic spread sheet modeling**
- **Understanding range names**
- **LOOKUP functions**
- **INDEX function**
- **MATCH function**
- **Text manipulation functions**
- **Time – Dates and date functions**
- **Conditional statements (IF statement)**



DA102 – Detailed contents

- **Three-dimensional formulas**
- **Sensitivity analysis**
- **COUNT family**
- **SUM family**
- **OFFSET function**
- **INDIRECT function**
- **Data validation**
- **Filtering and removing duplicates**
- **Consolidating data**

DA102 – Detailed contents



- **Pivot tables**